Midterm Report — RL-Assisted RAG: Reward Design

Course: MSML635 — Reinforcement Learning **Group:** 5 **Date:** 2026-10-20 (Week 2 of a 7-week project) **Final deadline:** End of November 2026 **Repository:** https://github.com/elderflamevandle/RL-MSML635-Group-Project

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Abstract

We are investigating **reward design for RL-assisted Retrieval-Augmented Generation (RAG)**. A RAG policy must decide, per query, whether to (a) take an `<evidence>` path and quote supporting sentences from retrieved passages, or (b) take an `<llm>` path and abstain when evidence is insufficient. Our baseline is the GRACE framework (ICLR 2025), which uses a summed reward R_0. We are testing two alternatives — $\mathbf{R_1}$ (cost-sensitive with a hallucination penalty) and $\mathbf{R_2}$ (hard-gated multiplicative) — against R_0 under both **GRPO** and **DAPO** policy-optimization recipes. Our research questions are **RQ1: does reward *gating* reduce hallucination on unanswerable queries?** and **RQ2: how do GRPO and DAPO compare in accuracy and sample efficiency on this task?**

At midterm, **all engineering infrastructure is complete** — rewards, retrievers, evaluation harness, caches, and 88 unit tests — and we have a validated **pre-RL baseline** on Qwen3-0.6B (Balanced Accuracy = 0.04 on HotpotQA tiny). **RL training itself is pending GPU-server provisioning**, which we expect to resolve this week and treat as our Week-2 gate.

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1. Introduction

1.1 Problem

Modern RAG systems either hallucinate when retrieval is weak (claiming answers that aren't in the passages) or over-abstain when retrieval is strong (refusing to answer well-supported questions). GRACE (ICLR 2025) tackles this with a per-query path choice — `<evidence>` for grounded answers, `<llm>` for honest abstention — and a reward that sums bonuses for correct path, format, and evidence quality.

1.2 Hypothesis

Summed rewards are susceptible to **ceiling asymmetry**: the `<evidence>` path's maximum reward (`R_f + R_p + α + β` ≈ 2.5) exceeds the `<llm>` path's maximum (`R_f + R_p` = 1.5), biasing the policy toward answering even when it should abstain. If we gate on each requirement multiplicatively, so a missing piece collapses the reward to 0 rather than merely reducing it, the policy should learn to abstain more honestly.

1.3 Research questions

- **RQ1.** Does gating the reward (vs summing all terms) stop the policy from guessing on insufficient evidence?
- **RQ2.** With the reward fixed, how do **GRPO** and **DAPO** compare in balanced accuracy and sample efficiency?

1.4 Deliverables

A **6-run experimental grid** (3 rewards × 2 optimizers) on HotpotQA, plus a stretch run on QASPER for cross-domain validation and a Qwen3-1.7B "hero" run on the winning combination.

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2. Approach

2.1 Reward designs

R_0 — GRACE baseline (control). Summed reward with branches on format validity and path choice:

```
R₀ = 0                                  if format invalid
   = R_f                                 if valid format, wrong path
   = R_f + R_p                           if correct <llm> path
   = R_f + R_p + α·R_Lev + β·R_Lans      if correct <evidence> path
```

with `R_f = 0.5`, `R_p = 1.0`, and `R_Lev / R_Lans` = ROUGE-L against gold evidence and gold answer respectively. α, β are inherited from GRACE's published values.

R_1 — Cost-sensitive with hallucination penalty. Adds (a) an explicit `-λ` penalty when the policy answers an unanswerable query, (b) a quality bonus `γ · R_Lans` on the `<llm>` path to close the ceiling asymmetry, and (c) a quote-containment indicator `𝟙[quote ⊂ C_ret]` that gates the format bonus on the `<evidence>` path — so fabricated evidence (ROUGE-matches gold but wasn't in the retrieved context) earns zero format credit.

R_2 — Hard-gated multiplicative (RQ1 extreme). All terms multiply:

```
R₂ = R_format × (w_p · R_path + w_q · R_path · R_quality)
```

where `R_format, R_path ∈ {0, 1}`. Any failure (invalid format, wrong path, non-verbatim quote) collapses R_2 to 0 — no partial credit. If R_2 wins on Balanced Accuracy, gating works; if R_2 collapses the policy to always-abstain, gating is too strict and R_1 is the sweet spot. Either outcome answers RQ1.

Starting hyperparameters: `λ = 1.0`, `γ = 0.5`, `w_p = 0.4`, `w_q = 0.6`. No sweeps until Week 6.

2.2 Policy optimization

Both **GRPO** (Group Relative Policy Optimization) and **DAPO** (Decoupled Clip and Dynamic sAmpling Policy Optimization) via `verl` at the GRACE paper's pinned commit `4aa02fe`. No custom algorithm code — just reward functions plugged into verl recipes.

2.3 Datasets

- **HotpotQA** (primary). 2000 train + 500 val from the `distractor` split, balanced ~50/50 answerable/unanswerable via a deterministic distractor-swap procedure (SHA-1 hash of `sample_id` seeded with target rate 0.45).
- **QASPER** (stretch, Week 5). Cross-domain validation on scientific papers; loader already implemented.

2.4 Model

Qwen3-0.6B + LoRA for the main grid (fits on a single 6 GB GPU); **Qwen3-1.7B hero run** on the winning (reward, optimizer) in Week 5.

2.5 Evaluation

- **Exact Match (EM)** on answerable queries — primary accuracy.
- **Abstention Accuracy (TNR)** on unanswerable queries.
- **Balanced Accuracy = mean(EM, Abstention)** — primary metric.
- **BERTScore** — semantic-similarity sanity check.
- **LLM-as-Judge** — Groq-hosted Llama-3.3-70B at temperature 0, with on-disk caching keyed by prompt+response hash. Validated for ≥85% agreement with EM on a 50-example hand-labeled subset.

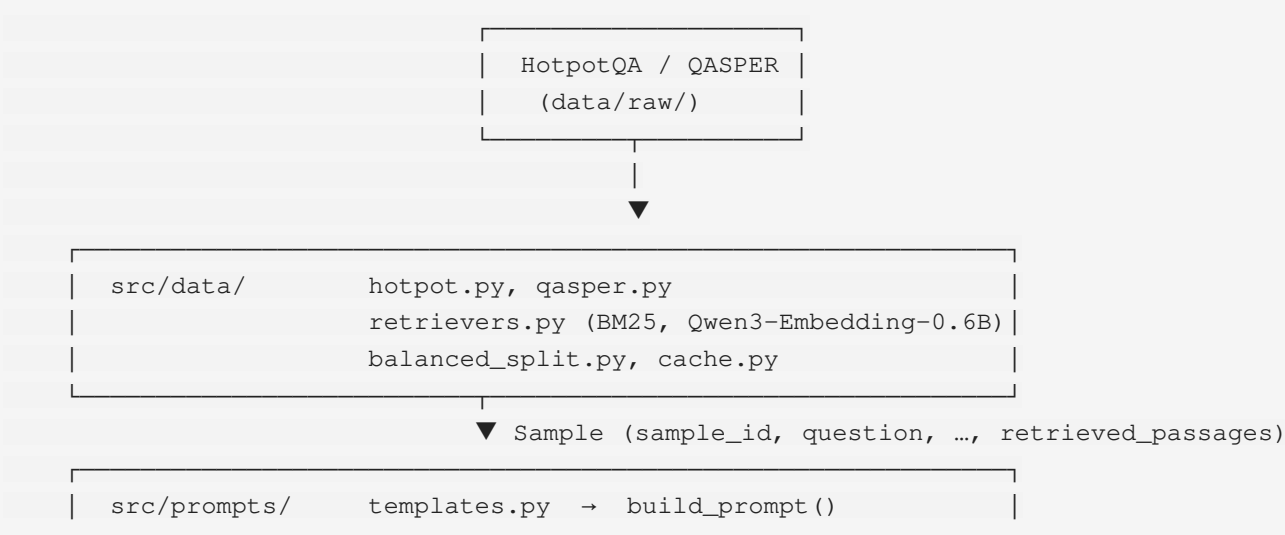
2.6 Experimental grid

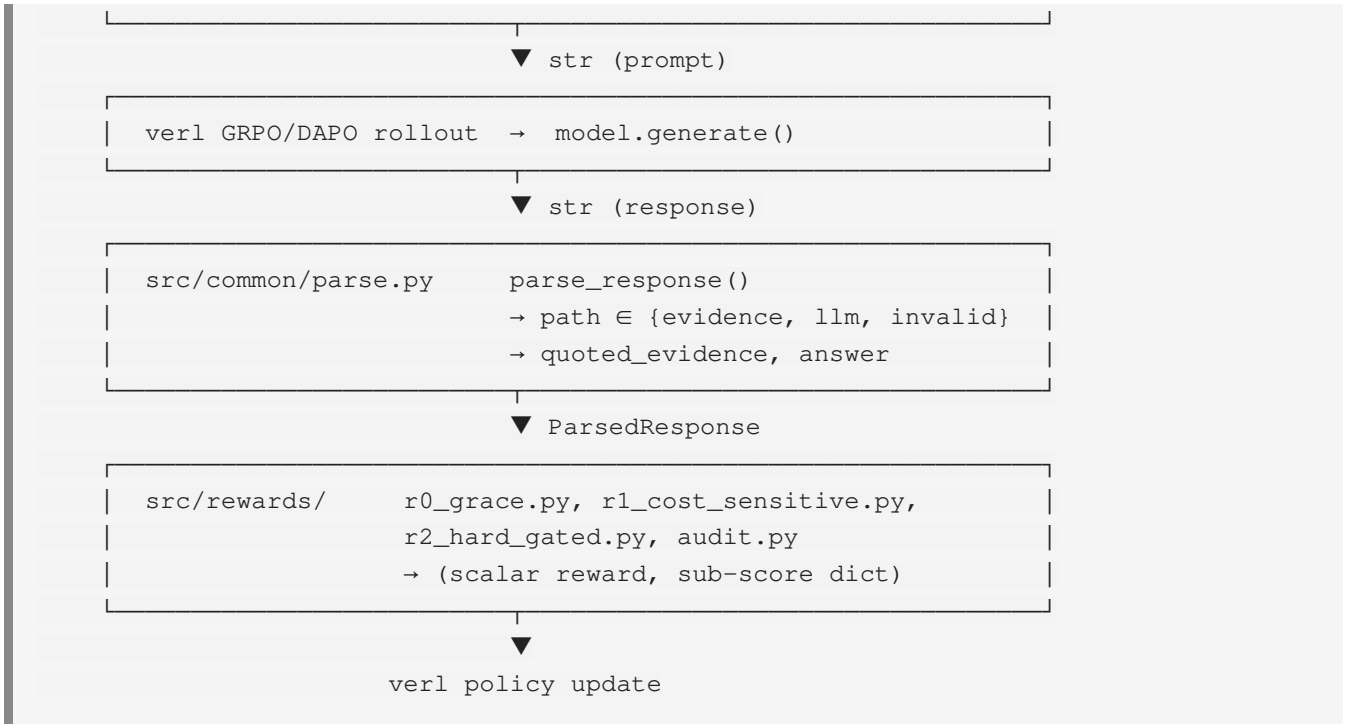
Reward	Optimizer	Dataset	Priority
R_0	GRPO	HotpotQA	**must** (baseline)
R_0	DAPO	HotpotQA	**must** (RQ2 half)
R_1	GRPO	HotpotQA	**must**
R_2	GRPO	HotpotQA	**must** (RQ1 answer)
R_1	DAPO	HotpotQA	should
R_2	DAPO	HotpotQA	should (completes RQ1 × RQ2)
Best(R*, opt) on Qwen3-1.7B	—	HotpotQA	should (hero)
Best(R*, opt)	—	QASPER	stretch (cross-domain)

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3. System implementation

3.1 Architecture overview





The **integration contract** is `Sample` (dataclass) + `parse_response()` (single-source-of-truth parser). All three member lanes compile against this contract, so each can be developed independently on a laptop.

3.2 What is implemented and tested (as of 2026-10-20)

Component	Files	Status
HotpotQA loader + balanced split	`src/data/hotpot.py`, `src/data/balanced_split.py`	✓ 3 tests
QASPER loader (cross-domain)	`src/data/qasper.py`	✓ 10 tests
BM25 retriever	`src/data/retrievers.py`	✓ 2 tests, 2000 train + 500 val built
Dense retriever (Qwen3-Embedding-0.6B)	`src/data/retrievers.py`	✓ 1 test, server-side build pending
Retrieval cache (SHA-256 keys, atomic, collision-safe)	`src/data/cache.py`	✓ 2 tests, 4000 entries cached
Prompt templates	`src/prompts/templates.py`	✓ 1 test + 3 round-trip tests with parser
Response parser	`src/common/parse.py`	✓ 5 tests
R_0 reward (GRACE control)	`src/rewards/r0_grace.py`	✓ 9 tests (all 4 branches + edge cases)
R_1 reward (cost-sensitive)	`src/rewards/r1_cost_sensitive.py`	✓ unit tests (part of 14 R1/R2/audit tests)
R_2 reward (hard-gated)	`src/rewards/r2_hard_gated.py`	✓ unit tests
ROUGE-L + quote-containment utilities	`src/rewards/metrics.py`	✓ 7 tests

Reward-hacking audit utilities	`src/rewards/audit.py`	✓ tests
Eval metrics (EM, abstention, balanced acc)	`src/eval/metrics.py`	✓ 6 tests
Eval harness (vLLM-backed)	`src/eval/evaluate.py`	✓ laptop-import-safe (lazy vLLM)
Local sanity runner (laptop GPU)	`src/eval/local_runner.py`, `scripts/local_sanity.py`	✓ verified end-to-end on 50 samples
Groq LLM-as-Judge client	`src/eval/judge.py`	✓ structure + caching done; needs key to validate
Metrics aggregation + comparison tables	`src/eval/aggregate.py`, `scripts/make_comparison_table.py`	✓ 3 tests
verl training launcher + run-naming	`scripts/launch_run.sh`, `src/training/run_naming.py`	✓ server-side, pinned to GRACE's verl commit `4aa02fe`
Base GRPO config	`configs/r0_grpo.yaml`, `configs/skeleton.yaml`	✓ draft; needs verl-field-name iteration on server

Test coverage: 88 passing (1 deselected slow test). All rewards, parser, retrievers, metrics, and data pipeline exercised. Determinism verified via stable argsort (retrievers) + SHA-256 cache keys.

3.3 Code artifacts worth noting

- **Collision-safe retrieval cache.** `src/data/cache.py` stores the full key tuple alongside the value and raises on hash collision, preventing silently corrupted retrievals across runs.
- **Fail-loud gold-passage assertion.** `src/data/hotpot.py` asserts non-empty gold passages at load time; silent breakage of ans/unans ratios is impossible.
- **Deterministic distractor swap.** `scripts/build_retrieval.py` uses `SHA-1(sample_id)` to pick ~45% of samples for forced-unanswerable, giving reproducible splits across machines.
- **Reward-value sanity check.** `0.1 < ans_rate < 0.95` assertion in the retrieval build fires if string normalization silently breaks the gold/retrieved match.

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4. Progress against plan

4.1 What is complete

Week	Task	Status
W0	Repo scaffold, dependency pin, test harness	✓
W1	Shared integration contract (`Sample`, `parse_response`)	✓
W1	Prompts + R_0 reward + ROUGE-L metrics	✓
W1	Training launcher + run naming + results convention	✓
W2	BM25 + dense retriever + balanced split + cache	✓ (BM25 built; dense pending server)
W2	R_0 edge-case tests + reward design doc	✓

W2	Eval harness skeleton (EM, abstention, balanced acc)	✓
W3	R_1 + R_2 + reward-hacking audit utilities	✓ (schedule-ahead)
W3	Groq LJ client structure	✓ (validation blocked on key)
W4	Metrics aggregation + comparison tables	✓ (schedule-ahead)
W5	QASPER loader	✓ (schedule-ahead)

Several Week 3-5 code deliverables have been **completed ahead of schedule** because they are laptop-safe (pure Python + unit tests) and do not depend on the GPU server. This gives us schedule buffer for the training runs themselves, which are the only remaining risk.

4.2 Pre-RL baseline (pipeline validation)

We ran the end-to-end pipeline against the **un-trained** Qwen3-0.6B base model on 50 HotpotQA samples (all answerable, from the tiny set). This is **not** an R0+GRPO result — it is a sanity check that the full chain (data → prompt → generate → parse → score) runs correctly, and a **floor** against which the Week-2 Gate (R0+GRPO) will be measured.

Metric	Value
Samples	50 (50 answerable / 0 unanswerable)
Inference time	25.5 s/sample (fp16 on RTX 4050 Laptop, 6 GB VRAM)
format_valid_rate	**0.70**
em_answerable	**0.080**
balanced_accuracy	**0.040**
path share: `<evidence>`	0.68
path share: `<llm>`	0.02
path share: `invalid`	0.30

Reading the baseline. The base model follows the prompt schema 70% of the time but gets the answer right only 8% of the time. When it does follow the schema, it routes to `<evidence>` almost exclusively — the `<llm>` abstention path barely exists untrained. **The failure mode most prevalent at the floor is invalid-format (30%), not wrong-path routing.** RL training's first measurable job is to drive the invalid share down and the `<llm>` share up (because our balanced split is ~50% unanswerable).

Gate target. The Week-2 Gate requires R0+GRPO to achieve **Balanced Accuracy ∈ [0.71, 0.91]** (within 10 points of GRACE's published HotpotQA number of 0.81). The gap from the current floor (0.04) is **~17×** — large but expected; this is the whole point of RL fine-tuning.

4.3 What is pending

Blocked on	Items
GPU server provisioning	Walking skeleton (1-step verl run); Week-2 Gate R0+GRPO run; full dense retriever build; all 6 grid runs; Qwen3-1.7B hero run
Groq API key	Judge validation (50-sample agreement check); re-eval of trained checkpoints with LJ

Team logistics decisions	Member assignments; results-sync mechanism; weekly meeting slot

The server provisioning is the critical path — nothing after Week 2 can begin until a walking-skeleton verl run completes on real hardware.

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5. Risks and mitigations

Risk	Likelihood	Impact	Mitigation
verl install / GPU driver issues	High	Blocks everything	3-day walking-skeleton pair-debug session; if still blocked, fallback to vanilla TRL + GRPO (loses DAPO, keeps RQ1)
R0+GRPO Balanced Accuracy wildly off GRACE's 0.81	Medium	Blocks R_1/R_2 interpretation	Hard gate forces whole-team halt; check prompt format, ans/unans ratio, LoRA rank, model match; 1 week of Week 3 budget reserved for this
R_2 collapses to always-abstain	Medium	Expected failure mode — **informative, not fatal**	Documented as a positive RQ1 result (gating too strong); add R1.5 with soft gates if time permits
DAPO config instability	Medium	Weakens RQ2 answer	Use verl's default DAPO hyperparams from GRACE recipe verbatim — do not tune (would swamp the GRPO/DAPO signal)
Groq rate limit mid-eval	Low	Eval delays	On-disk cache + sleep-and-retry on 429; fallback to local Qwen3-4B zero-shot judge
Member unavailable for a week	Medium	Lane stalls	Shared `Sample` + `parse_response` contract means another member can pick up the lane; Member 3 can run Member 1's eval; Member 1 can run Member 2's audits
Qwen3-0.6B ceiling too low to show reward differences	Low-Med	Weak final signal	Week-5 Qwen3-1.7B hero run exists for exactly this; grid is kept cheap enough to afford it
HF dataset format drift	Low	Day-1 blocker	HuggingFace datasets revision hash pinned

5.1 Known stack issues (new since proposal)

- **Local dev stack misalignment.** `sentence-transformers` requires `torch ≥ 2.6` (torch.accelerator API), but our pinned stack is `torch 2.5.1+cu121`. Workaround: pinned `transformers<5`, `datasets<5`, `sentence-transformers<5`; slow dense-retriever test deselected. **Needs resolution before server deployment** — team decision in Week 3 on upgrade vs stay-pinned.
- **Windows system Python 3.14 shadowing.** On the dev laptop, `python` resolves to `%APPDATA%\Python\Python314` (CPU-only torch 2.11), not the `.venv` (torch 2.5+cu121). Documented in `docs/log/2026-10-20-local-sanity.md`; teammates on macOS/Linux will not hit this.

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6. Plan for the remainder of the project

Overall approach. A **3-day whole-team walking-skeleton debug** on the server, then a fan-out: Member 1 owns data/eval, Member 2 owns rewards/audit, Member 3 owns training/configs. All three converge again for the Week-5 report.

6.1 Week-by-week

Week 3 (Oct 26 – Nov 1) — Walking skeleton & R_0 reproduction. Server env setup → iterate on verl YAML field names until 1 training step completes → launch the full R0+GRPO run → first evaluation. Member 1 wires up Groq LJ + 50-example hand-labeled validation. **Gate: R0+GRPO Balanced Accuracy $\in$ [0.71, 0.91].**

Week 4 (Nov 2 – 8) — DAPO + full "must" grid + first analysis. R0+DAPO config; launch all 4 must-have runs; Member 2 runs reward-hacking audits on completed checkpoints and iterates R_1/R_2 **only if audits show real problems**; Member 1 drafts aggregation tables. **Gate: comparison table with preliminary RQ1 and RQ2 signal.**

Week 5 (Nov 9 – 15) — Should runs + hero + report draft (target completion). R1+DAPO, R2+DAPO; Qwen3-1.7B hero run on the winning (reward, optimizer); report drafts of Abstract, Background, Method, Results, Discussion. **Gate: 6-run grid in one table + hero metrics + report drafted.**

Week 6 (Nov 16 – 22) — Buffer / ablations. If gates passed: α/β/γ/λ ablations on the winning reward; full QASPER cross-domain run; 20 failure cases per reward classified by failure type. If gates slipped: catch-up.

Week 7 (Nov 23 – 29) — Polish + submit. Figures, citations, final tables, presentation dry-run.

6.2 Success criteria

- **Minimum viable (must, by Week 5):** R_0 reproduction within 10 pts of GRACE + $R_0/R_1/R_2$ × GRPO on HotpotQA + one comparison table + RQ1 answered with evidence.
- **Full success (target, by Week 5):** Minimum viable + full 6-run grid + per-run reward-hacking audit + RQ2 answered via GRPO vs DAPO sample-efficiency curves + Qwen3-1.7B hero.
- **Stretch (Weeks 6-7):** QASPER cross-domain + hyperparameter ablation on winner + error-case classification.

6.3 What "answered RQ1 / RQ2" looks like

- **RQ1 answered** when Balanced Accuracy and unanswerable abstention accuracy for $R_0/R_1/R_2$ are plotted side by side. A monotonic improvement $R_0 \rightarrow R_1 \rightarrow R_2$, or an informative inversion at R_2, is the result.
- **RQ2 answered** when GRPO and DAPO sample-efficiency curves (Balanced Accuracy vs training step) are overlaid for the same reward. The faster-converging / higher-plateau optimizer is the answer.

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7. Team organization

Member	Lane	Primary files

Member 1 (Data)	HotpotQA/QASPER loaders, retrievers, cache, eval infra, LJ validation	`src/data/*`, `src/eval/*`
Member 2 (Rewards)	$R_0/R_1/R_2$ implementation, reward-hacking audits, reward-design report section	`src/rewards/*`, `scripts/run_audits.py`, `docs/rewards.md`
Member 3 (Training)	verl env, launchers, GRPO/DAPO configs, grid runner, hero run, training comparison report section	`scripts/setup_env.sh`, `scripts/launch_run.sh`, `configs/*.yaml`, `src/training/*`

Day-1 team-logistics checklist in `docs/team.md` (committed, drafted with proposed defaults for review).

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8. Reproducibility

Everything a reviewer needs to reproduce midterm numbers:

```
# Clone + install (Python 3.12, CPU wheels OK for pipeline; CUDA 12.1 for GPU inference)
git clone https://github.com/elderflamevandle/RL-MSML635-Group-Project.git
cd RL-MSML635-Group-Project
python -m venv .venv && source .venv/bin/activate  # or .venv/Scripts/activate on Window
pip install -e ".[dev]"

# Data
python scripts/download_hotpot_tiny.py            # 50-sample tiny set
python scripts/download_hotpot_full.py             # 2000 train + 500 val
python scripts/build_retrieval.py --no-dense       # BM25-only; dense requires torch ≥ 2

# Tests
pytest -q                                          # expect "88 passed, 1 deselected"

# Pre-RL baseline (requires ~5 GB VRAM, ~20 min on RTX 4050)
python scripts/local_sanity.py --n 50 --device cuda
# Writes results/local_sanity/<timestamp>/metrics.json
```

Determinism knobs: HuggingFace dataset revision pinned; retrieval cache keyed by SHA-256; argsort used in retriever ranking is stable; distractor-swap seeded by SHA-1(`sample_id`).

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9. Conclusion

At the midterm mark we have delivered **all code infrastructure** — three reward designs, both algorithm configurations, HotpotQA and QASPER data loaders, retrievers (BM25 shipped, dense pending), evaluation harness, aggregation utilities, and 88 passing unit tests. We have validated the **pipeline end-to-end on real hardware** (Qwen3-0.6B on an RTX 4050, Balanced Accuracy = 0.04 as the pre-RL floor). The one open-ended risk is **GPU-server provisioning for the training runs themselves**, which is the sole blocker on the critical path from here through Week 5.

Our **success criteria are explicit and gated per week** (Section 6.2), and the plan has a **1-week slack in Week 3** reserved for the two highest-likelihood risks (verl install issues, GRACE-reproduction mismatch). The final report (end of November) will target the full 6-run grid, the Qwen3-1.7B hero run, and a reward-hacking audit per run.

References

1. GRACE: *Grounded Response and Correct Abstention via Reinforcement-Learned Policy Routing* — ICLR 2025 (reward R_0 design, verl commit `4aa02fe`).
2. Yang, Z., et al. *HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering*. EMNLP 2018.
3. Dasigi, P., et al. *A Dataset of Information-Seeking Questions and Answers Anchored in Research Papers* (QASPER). NAACL 2021.
4. Yu, Q., et al. *DAPO: An Open-Source LLM Reinforcement Learning System at Scale*. 2024.
5. Shao, Z., et al. *DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models* (GRPO). 2024.
6. Qwen Team. *Qwen3 Technical Report*. 2025.
7. Zhang, T., et al. *BERTScore: Evaluating Text Generation with BERT*. ICLR 2020.
8. verl framework — https://github.com/volcengine/verl

Appendices in-repo (not duplicated here):

- `docs/STATUS.md` — rolling status snapshot
- `docs/rewards.md` — reward implementation reference
- `docs/log/2026-10-20-local-sanity.md` — full baseline run log
- `docs/team.md` — team logistics
- `docs/superpowers/plans/2026-10-11-rl-rag-reward-design.md` — detailed task-level plan
- `HANDOFF.md` — Day-1 scaffold handoff
- `RL_Group_5_Proposal.pdf` — original project proposal